

Human-like Object Grouping in Self-supervised Vision Transformers

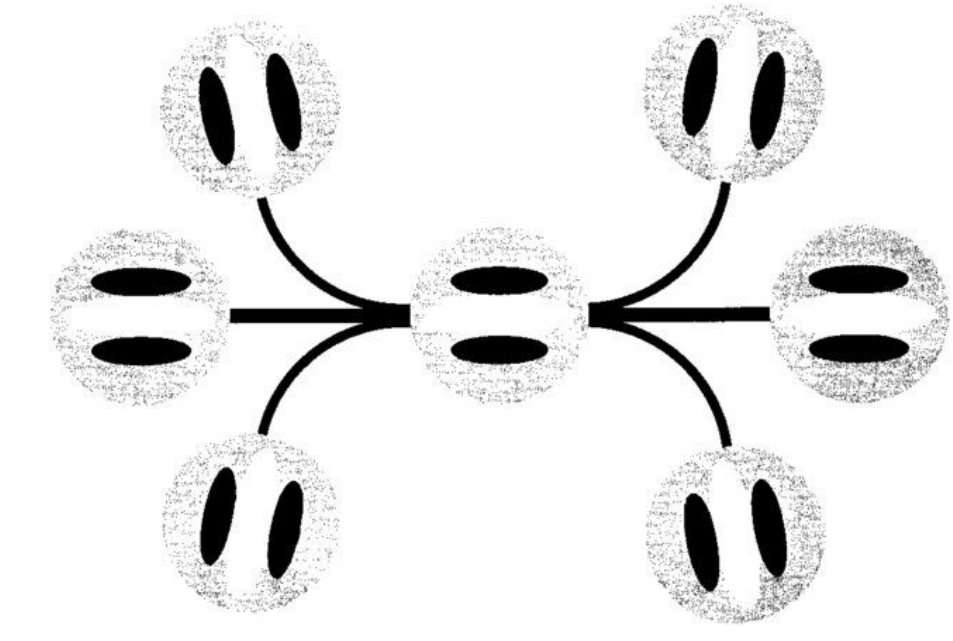
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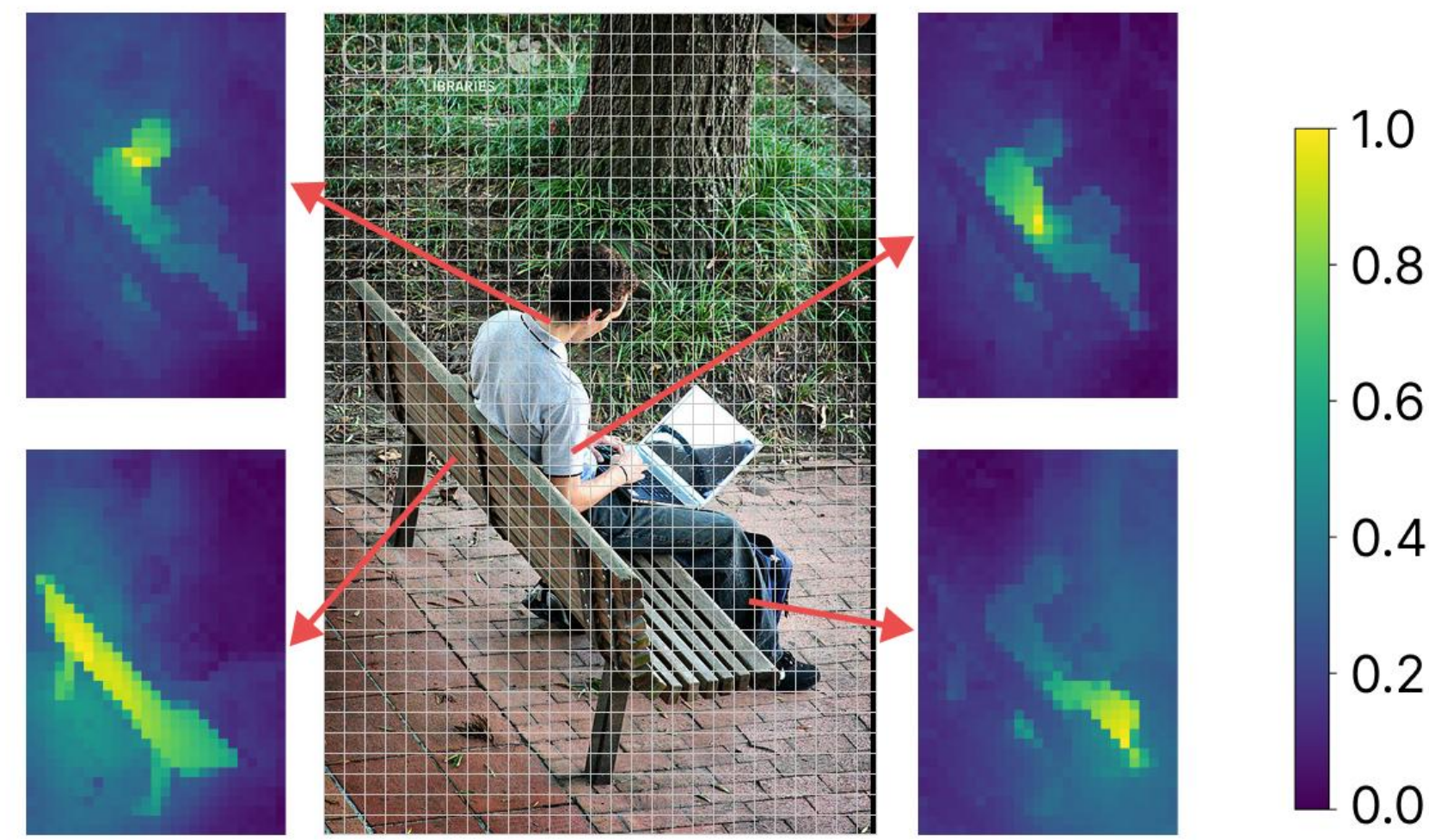


Association Fields in V1

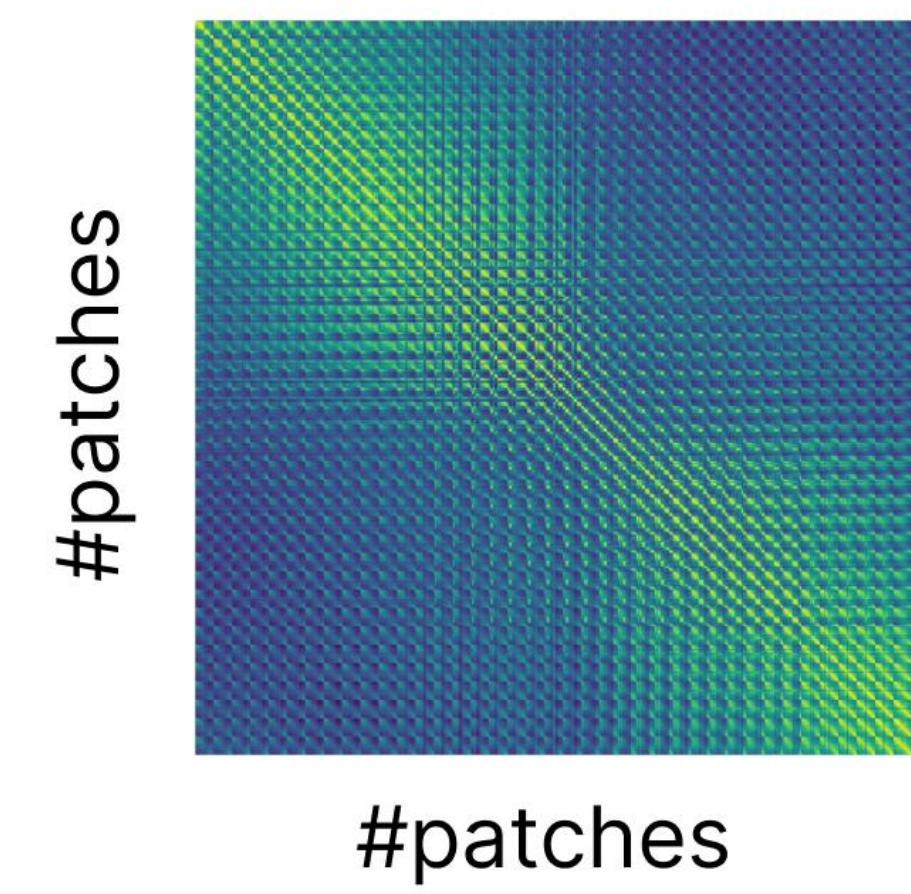


Field et al, (1993) [1]

Affinity maps



Gram matrix

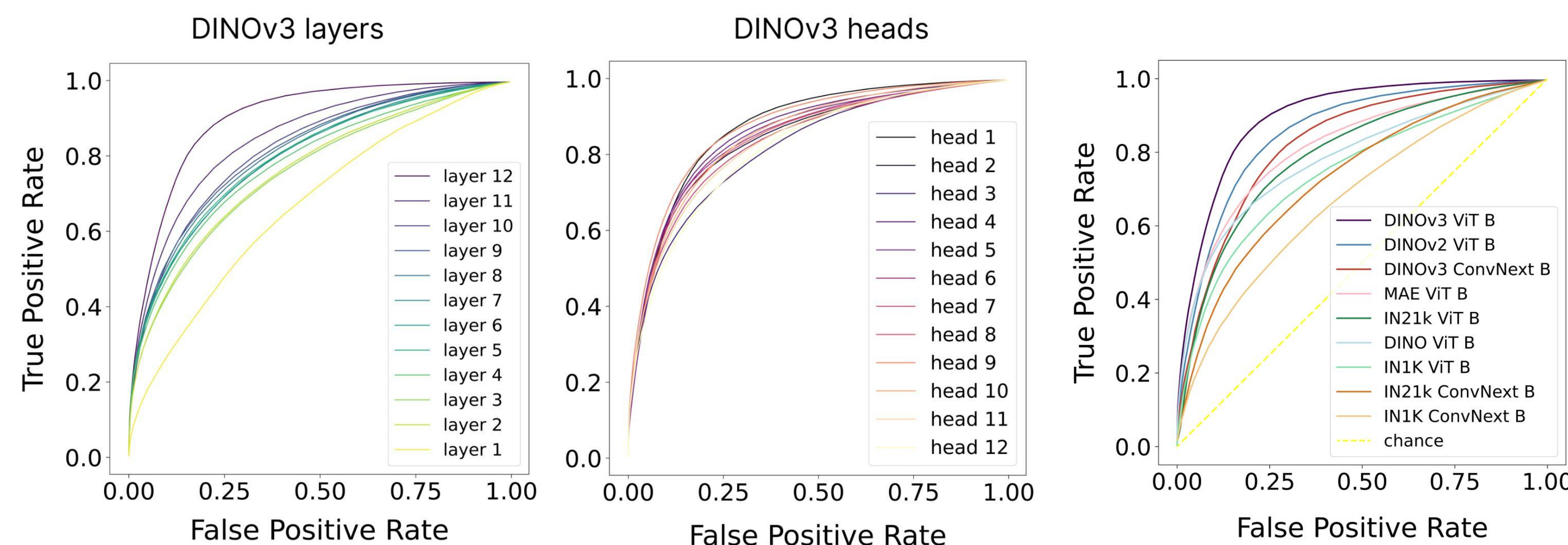
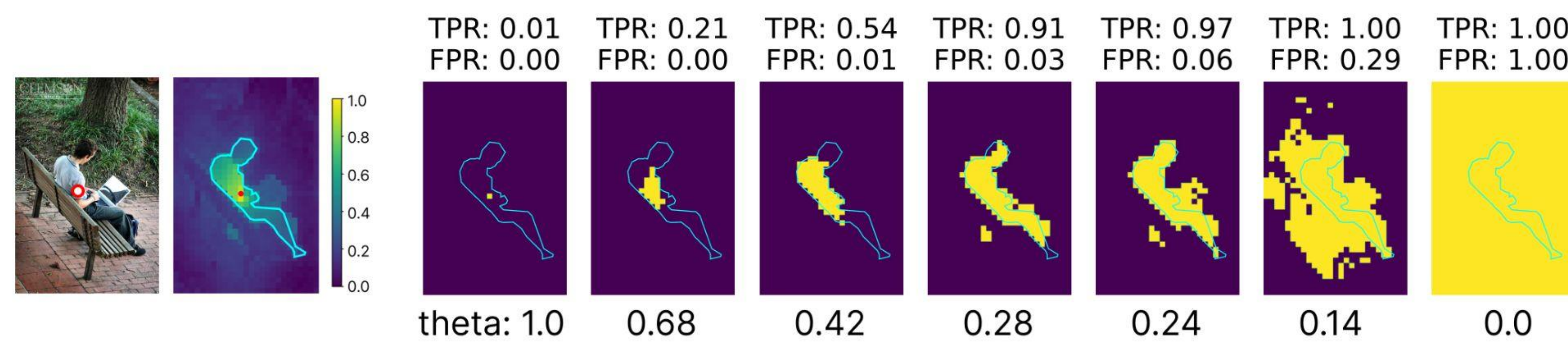


DINOv3 model [2]

- Object grouping in natural scenes requires the visual system **to bind image regions into coherent wholes** despite ambiguous boundaries and overlapping objects [3].
- Association fields [1] in early visual areas link neurons with similar low-level features, but grouping in natural scenes requires richer, object-centric connectivity in higher visual areas [4].
- Here we probe different vision models for how object-centric their representations are and how well they align with the grouping behavior of humans.

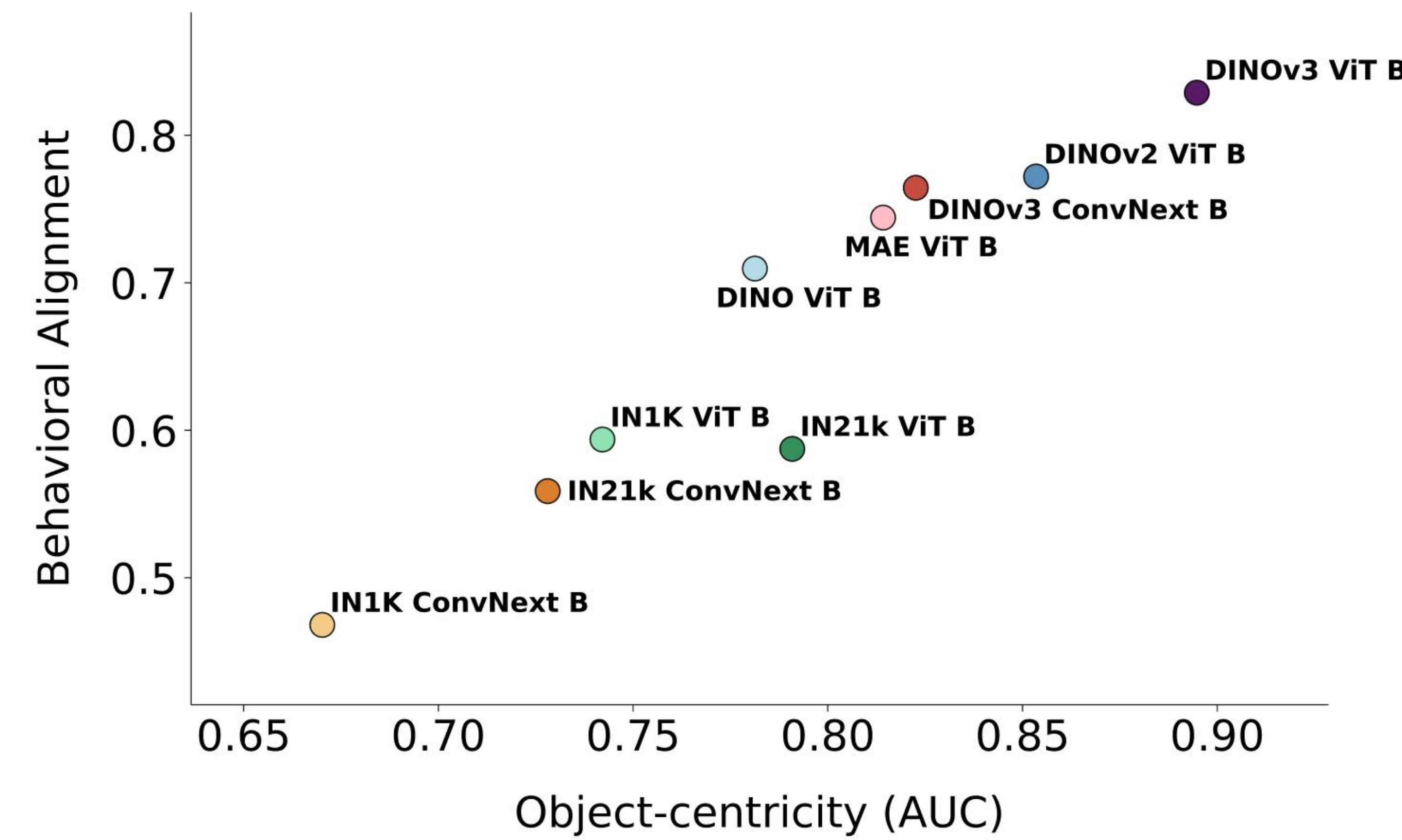
Object-centric Representations

For each image, we compute an **affinity map by taking the cosine similarity between the patch feature at the reference dot location and all other patch features**. We then perform an ROC analysis: TPR is the proportion of within-object patches whose affinity exceeds a threshold, FPR is the proportion of outside-object patches. Averaging across all trials yields a summary ROC curve per model, and we use the AUC as our object-centricity metric.

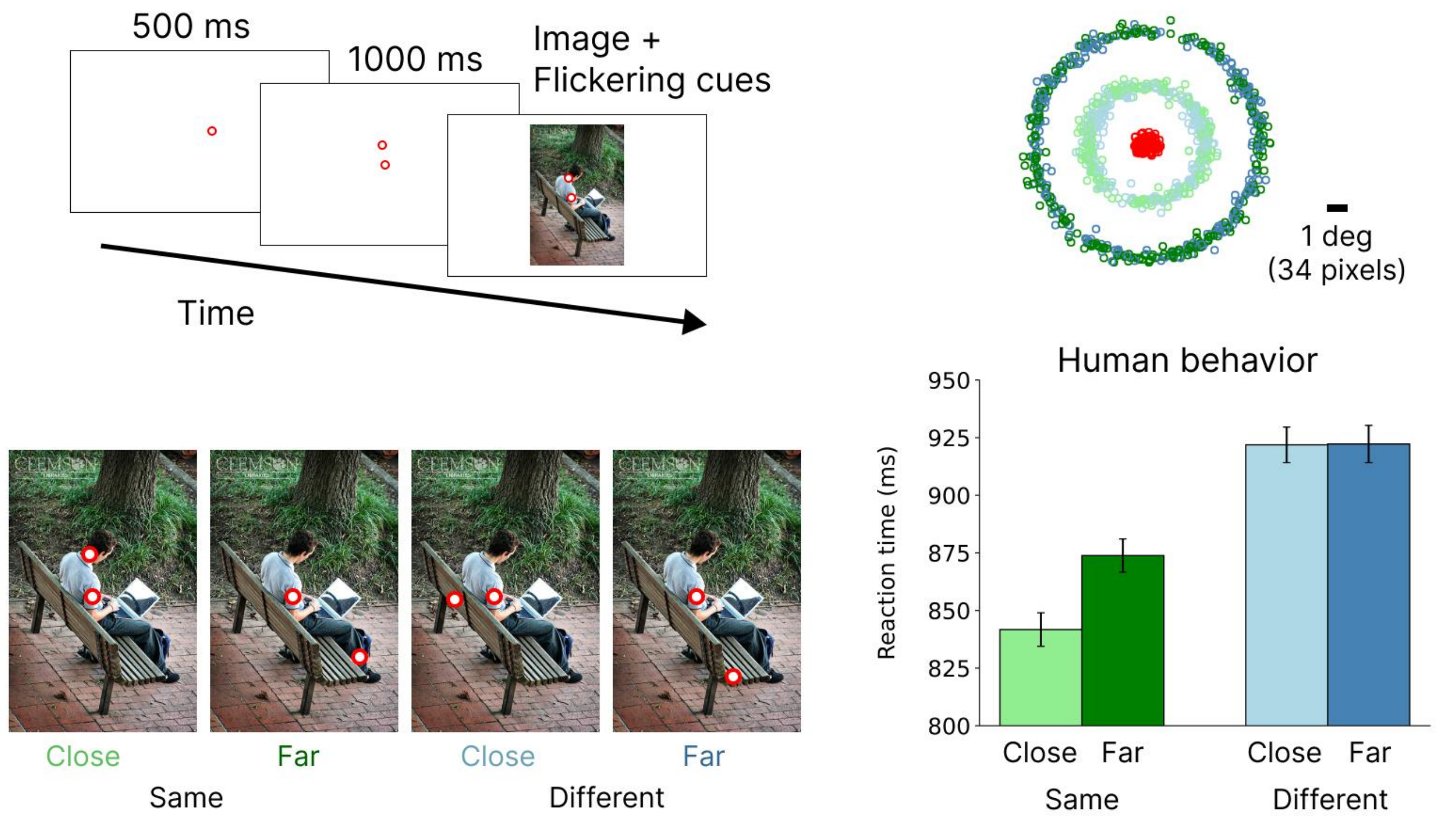


Behavioral Alignment

- Two-dot grouping task:** participants judged whether two dots fell on the same or different objects in natural scenes.
- 72 participants, 255 trials per participant, 1,020 unique trials across 255 images.
- Faster responses for same-object pairs; RT increased with dot distance only for same-object trials.

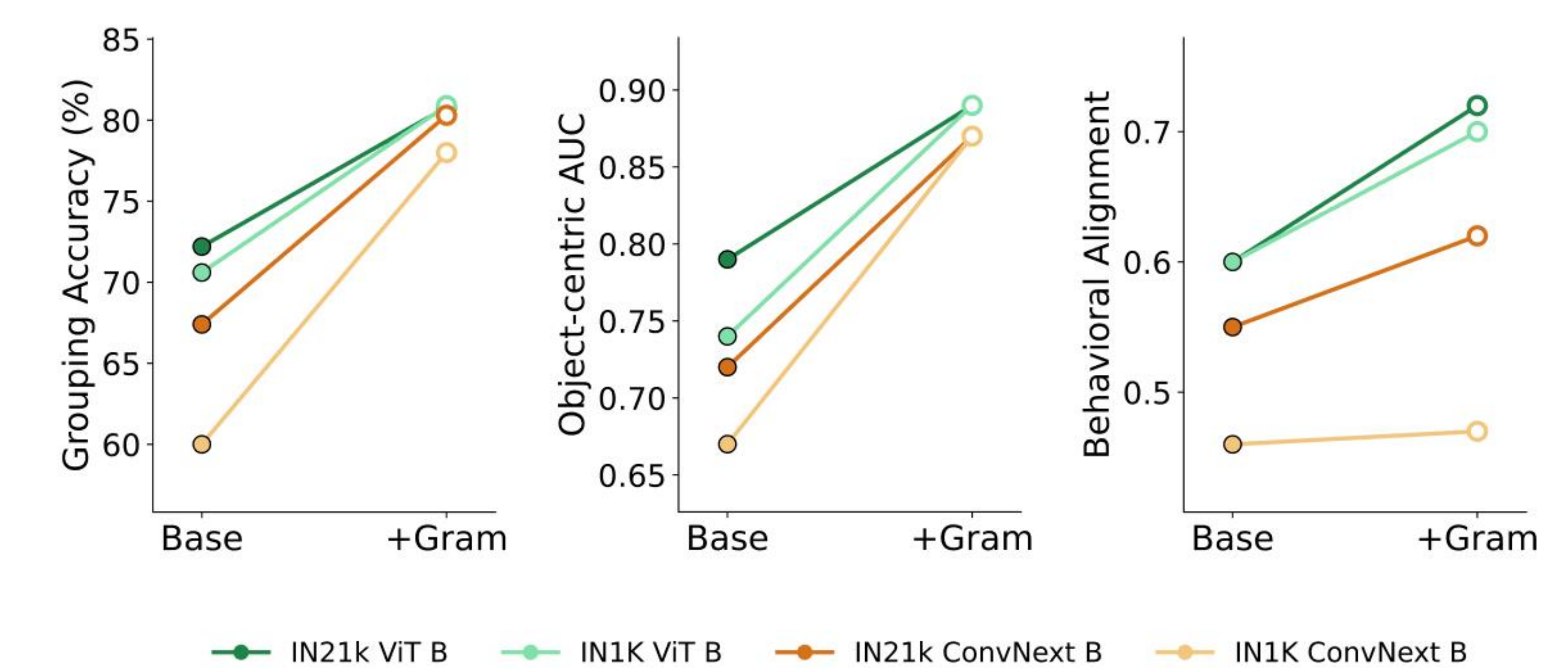


- Supervised ViT-B and ConvNeXt-B models are fine-tuned to match the DINOv3 ViT-B patch-level Gram matrix (pairwise similarity between all image patches).
- This improves grouping accuracy, object-centric AUC, and behavioral alignment across all four models.
- CLS-token distillation shows no improvement, highlighting Gram structure as the key factor.
- Transformers benefit more than CNNs, possibly because self-attention already models pairwise patch relationships.



- Patch features at the two dot locations predict same/different judgments and trial-level RTs using a simple MLP.
- Behavioral alignment** is noise-normalized Spearman correlation between predicted and human RTs.
- Object-centric AUC strongly predicts behavioral alignment across all 9 models ($r_s=0.950$, $p=0.0001$), spanning both transformers and CNNs.
- Stronger object-centricity of features is associated with more human-like grouping behavior.

Effect of Gram alignment



References

- [1] Field, D.J., Hayes, A., Hess, R.F.: Contour integration by the human visual system: evidence for a local "association field". Vision research 33(2), 173–193 (1993)
- [2] Siméoni, O., Vo, H.V., Seitzer, M., Baldassarre, F., Oquab, M., Jose, C., Khalidov, V., Szafraniec, M., Yi, S., Ramamonjisoa, M., et al.: Dinov3. arXiv preprint arXiv:2508.10104 (2025)
- [3] Roelfsema, P.R.: Solving the binding problem: Assemblies form when neurons enhance their firing rate—they don't need to oscillate or synchronize. Neuron 111(7), 1003–1019 (2023)
- [4] Gilbert, C.D., Li, W.: Top-down influences on visual processing. Nature reviews neuroscience 14(5), 350–363 (2013)

Paper



Poster

